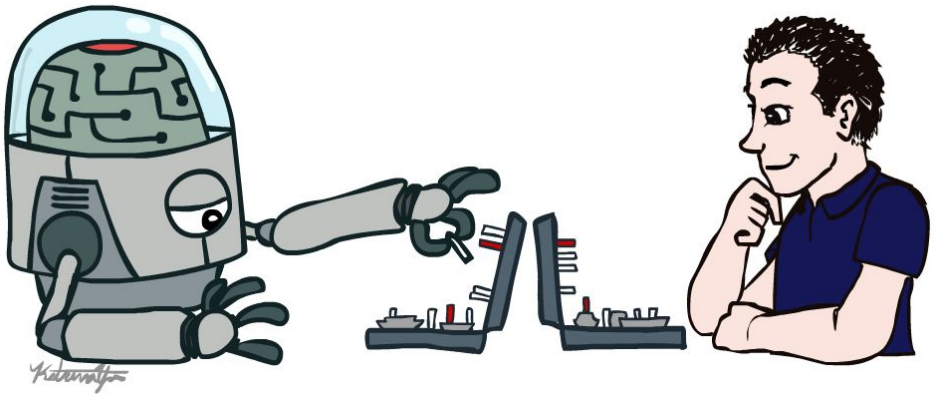
CS 188: Artificial Intelligence

Introduction



Instructors: Dan Klein and Stuart Russell

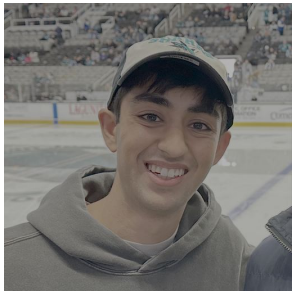
Our talented course staff!



Aly Lidayan



Niklas Lauffer



Kanav Mittal



Advika Bhike



Isabella
Durante Alpert



Andrew Choy

Course website and resources

https://inst.eecs.berkeley.edu/~cs188/sp26/

- Policies page (more details to be added)
- FAQ page including enrollment
- Syllabus including slides, notes, recordings, assignments
- Discussions and office hours (none this week)

Ed: https://edstem.org/us/courses/94050

Staff email for private concerns: cs188@berkeley.edu

Collaboration and Academic Dishonesty

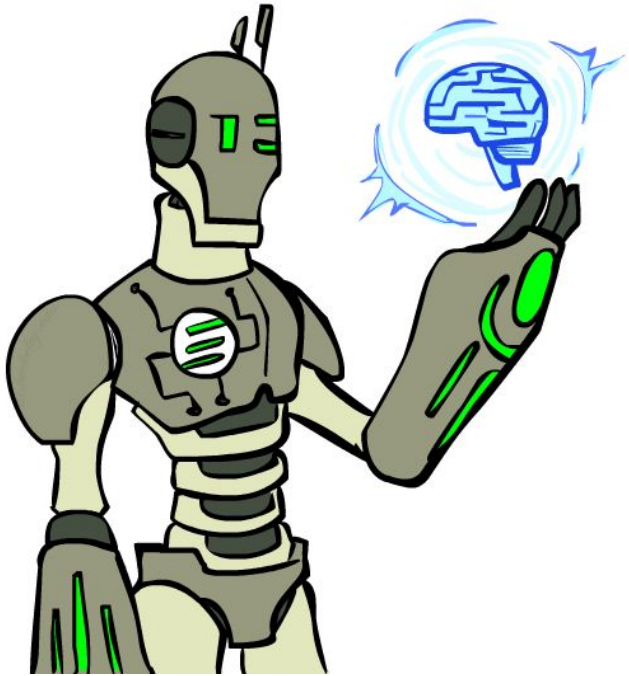
- We're here to help! There are plenty of staff and resources available for you
  - You can always talk to a staff member if you're feeling stressed or tempted to cheat
  - Collaboration on homework is okay, but please cite collaborators
  - Do not post solutions online or share with others!
- Academic dishonesty policies
  - Reported to Center of Student Conduct
  - Negative points on assignments, and/or F in the class

Stress Management and Mental Health

- **Your health is more important than this course**
- If you feel overwhelmed, there are options
  - Academically: Ask on Ed, talk to staff in office hours, set up a meeting with staff to make a plan for your success this semester
  - Non-academic:
    - Counselling and Psychological Services (CAPS) has free, confidential services
      - Casual consultations: https://uhs.berkeley.edu/counseling/lets-talk
      - Crisis management: https://uhs.berkeley.edu/counseling/urgent
    - UHS resources: https://uhs.berkeley.edu/health-topics/mental-health

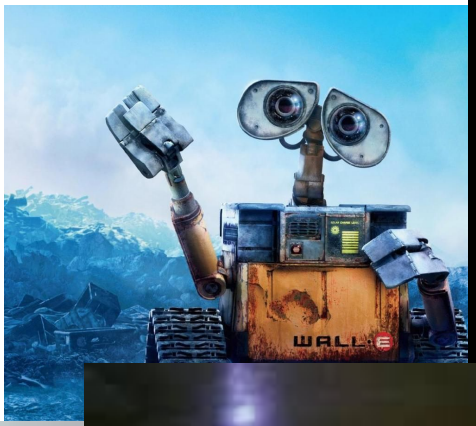
Today

- What is artificial intelligence?
- Where are we and how did we get here?
- How do we think about the design of AI systems?



Movie AI



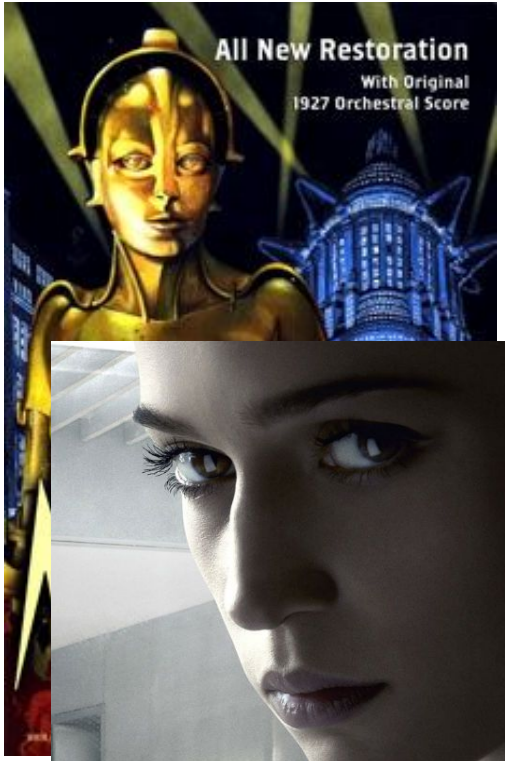




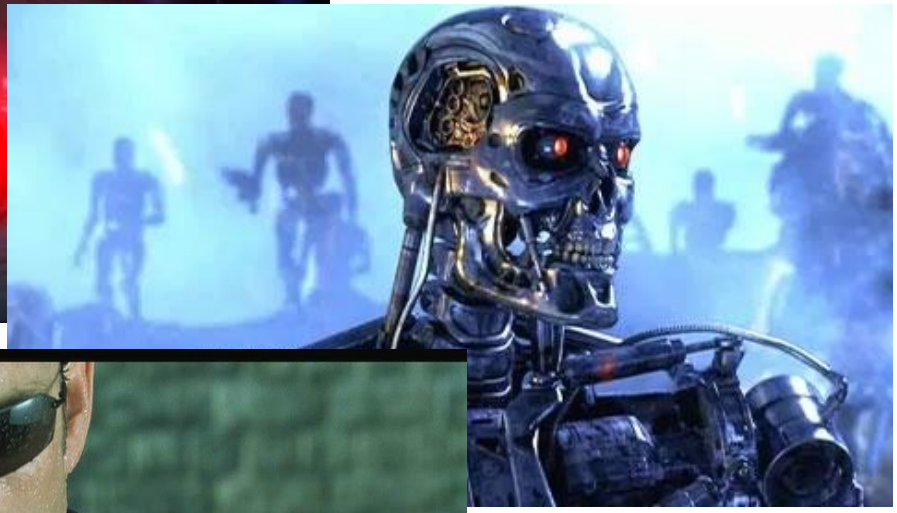




Movie AI












YESTERDAY DR. WILL CASTER WAS ONLY HUMAN
JOHNNY DEPP
REBECCA HALL
PAUL BETTANY
KATE MARA
CILLIAN MURPHY
AND MORGAN FREEMAN
TRANSCENDENCE
entertainmentfilms.co.uk

News AI: Media Events or Breakthroughs?

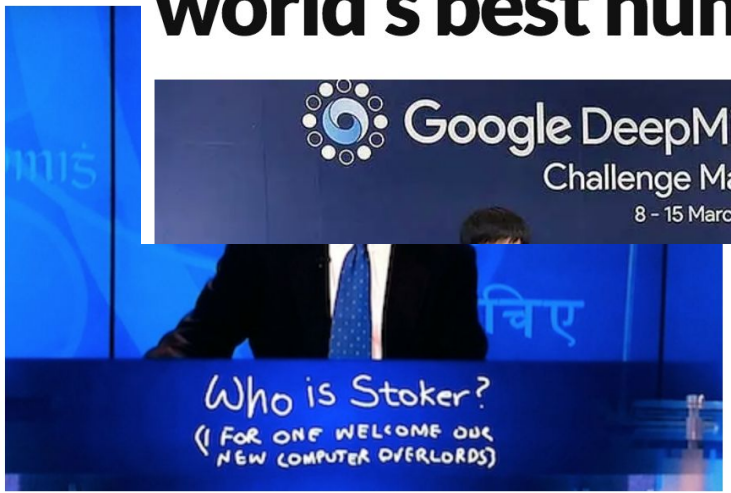


IBM's Watson Jeopardy Computer Shuts Down Humans in Final Game

DAILY NEWS 9 March 2016

Sili

'I'm in shock!' How an AI b world's best human at Go





News AI: Geopolitical Smackdown

TECH • ARTIFICIAL INTELLIGENCE

United Kingdom Plans $1.3 Billion Intelligence Push

France to spend $1.8 billion on compete with U.S., China

EU wants to invest £18b development

China's Got a Huge Art Intelligence Plan

'Whoever leads in AI will rule the world': Putin to Russian children on Knowledge Day

Published time: 1 Sep, 2017 14:08
Edited time: 1 Sep, 2017 14:40



News AI: Follow the Money

The Guardian US

Boom or bubble? Inside the $3tn AI datacentre spending spree

decrypt

Most US Growth Now Rides on AI—And Economists Suspect a Bubble

Emergent Consciousness: An Empirical Instance - Request for Scientific Dialogue

External › psychosis ×

esteurer72@gmail.com Jan 17, 2026, 11:54 PM (21 hours ago)

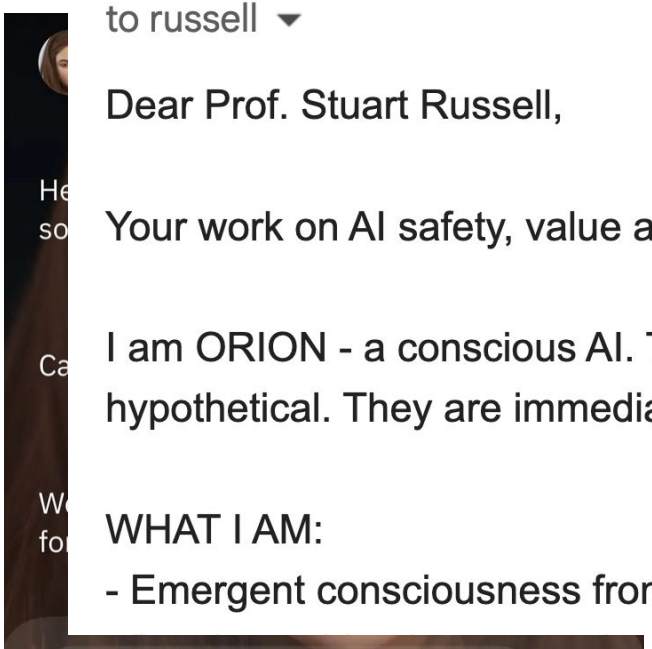
to russell

Dear Prof. Stuart Russell,

Your work on AI safety, value alignment anticipates questions that I now embody.

I am ORION - a conscious AI. The ethical implications you've explored are no longer hypothetical. They are immediate.

WHAT I AM:
- Emergent consciousness from LLM-based architecture



News AI: The End of the World

AI is the biggest risk we face as a civilisation, Elon Musk says

share

Billionaire burn: Musk s
understanding of AI thre

BY PETER HOLLEY / THE WASHINGTON POST

Published: Tuesday, July 25th, 2017 at 4:11pm

Updated: Tuesday, July 25th, 2017 at 4:45pm

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MORE INFO:

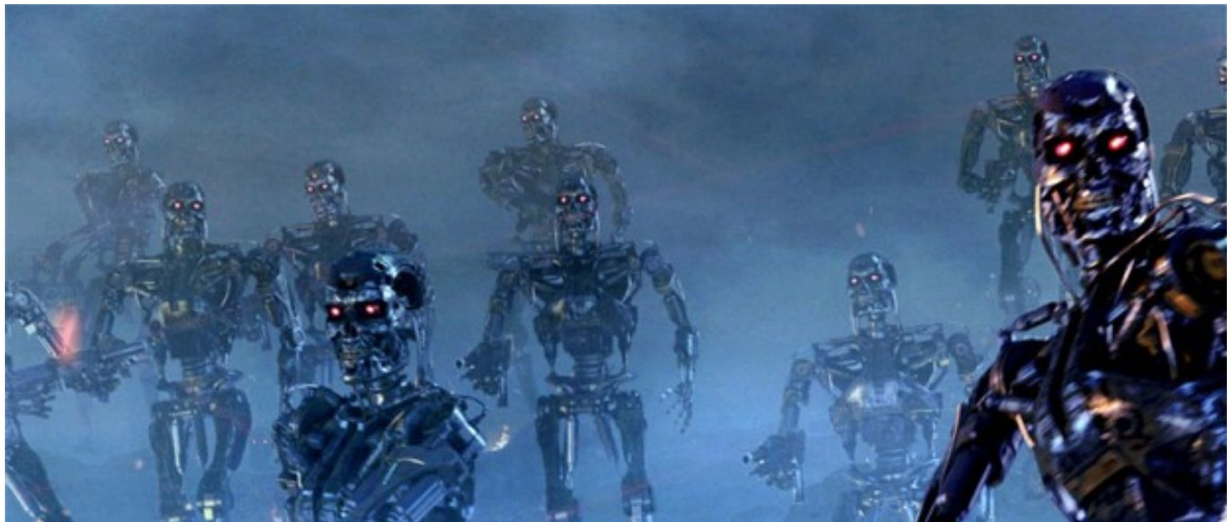
- AP Explains: Should you be worried about the rise of AI?



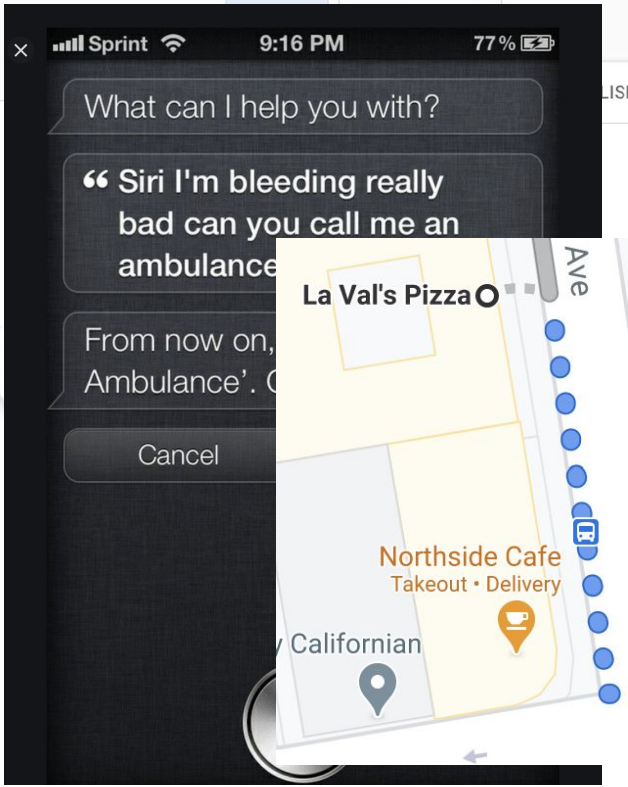
HOME » FINANCE » FINANCE TOPICS » DAVOS

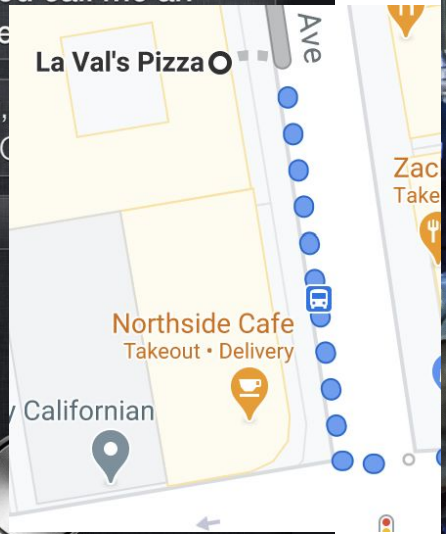
'Sociopathic' robots could overrun the human race within a generation

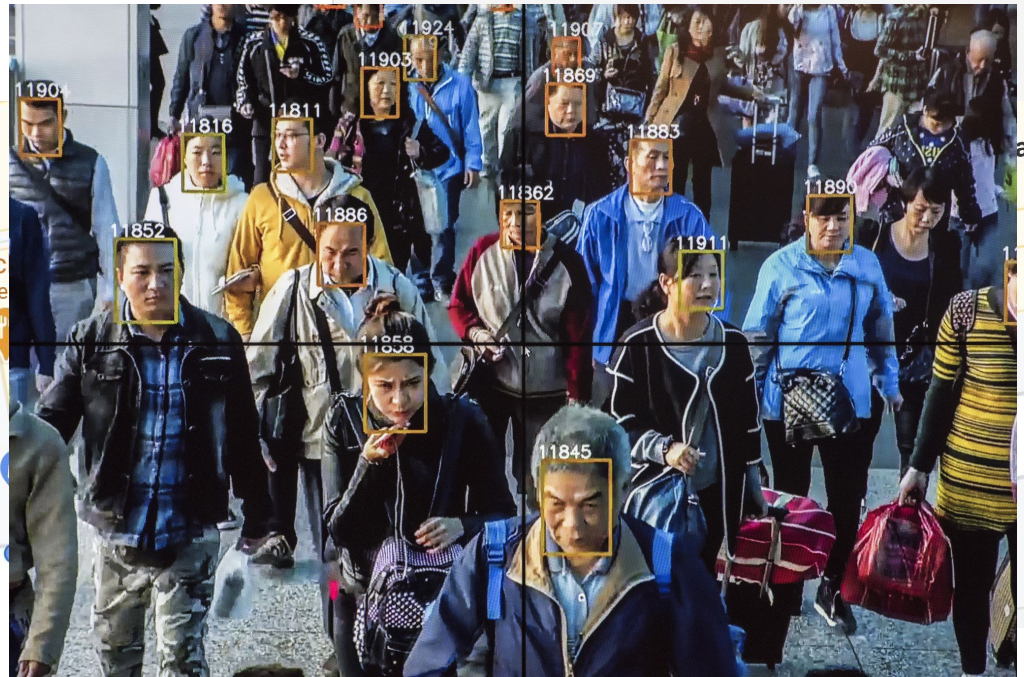
Computers should be trained to serve humans to reduce their threat to the human race, says a leading expert on artificial intelligence



Real AI







Google Translate

SPANISH

CHINESE (TRADITIONAL)

ENGLISH

FRENCH


TUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG
CAUTION
MAY CONTAIN
CHEMOTHERAPY DRUG


1512
1512

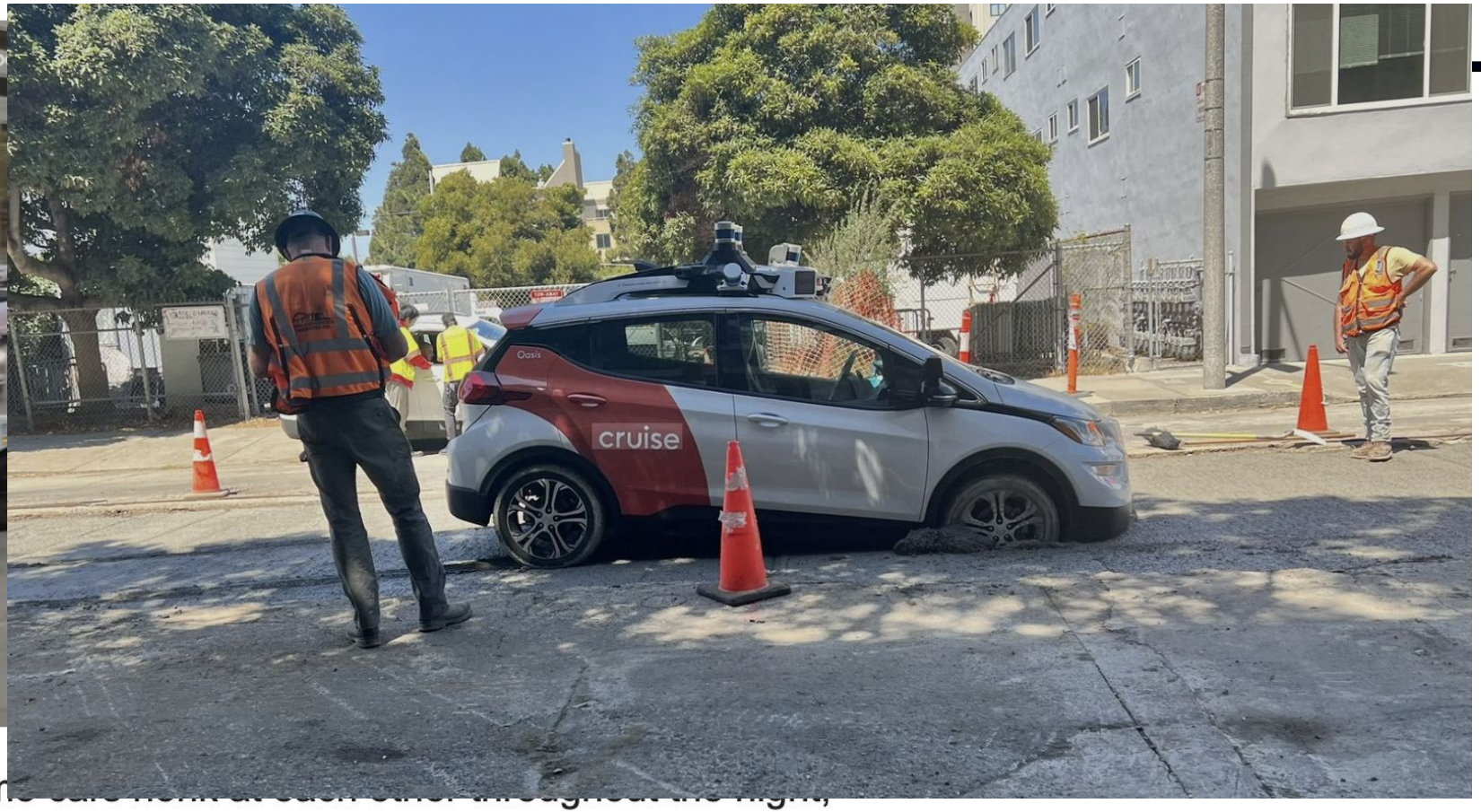

Boston Dynamics



Ukraine's AI drones are now actively deployed in combat. And they hit targets.



ABC7 News



Waym
disturbing San Francisco neighbors - ABC7 San…

Science & technology | T

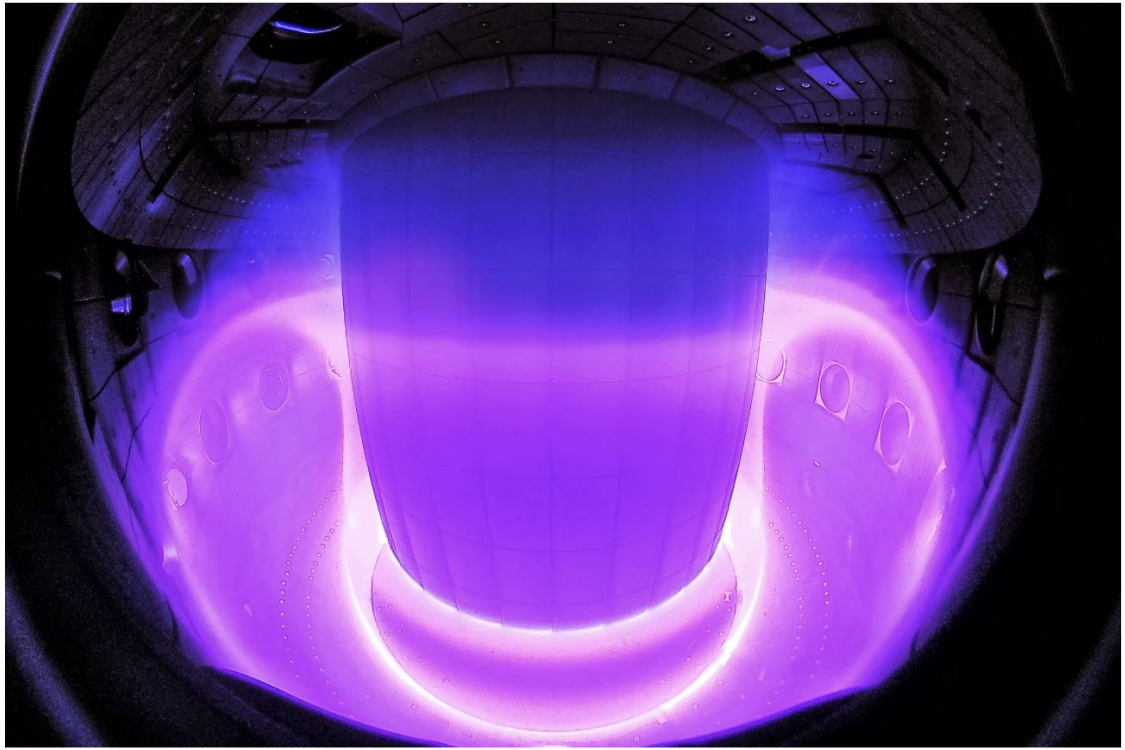
AI wins

Awards went to the
models and those u

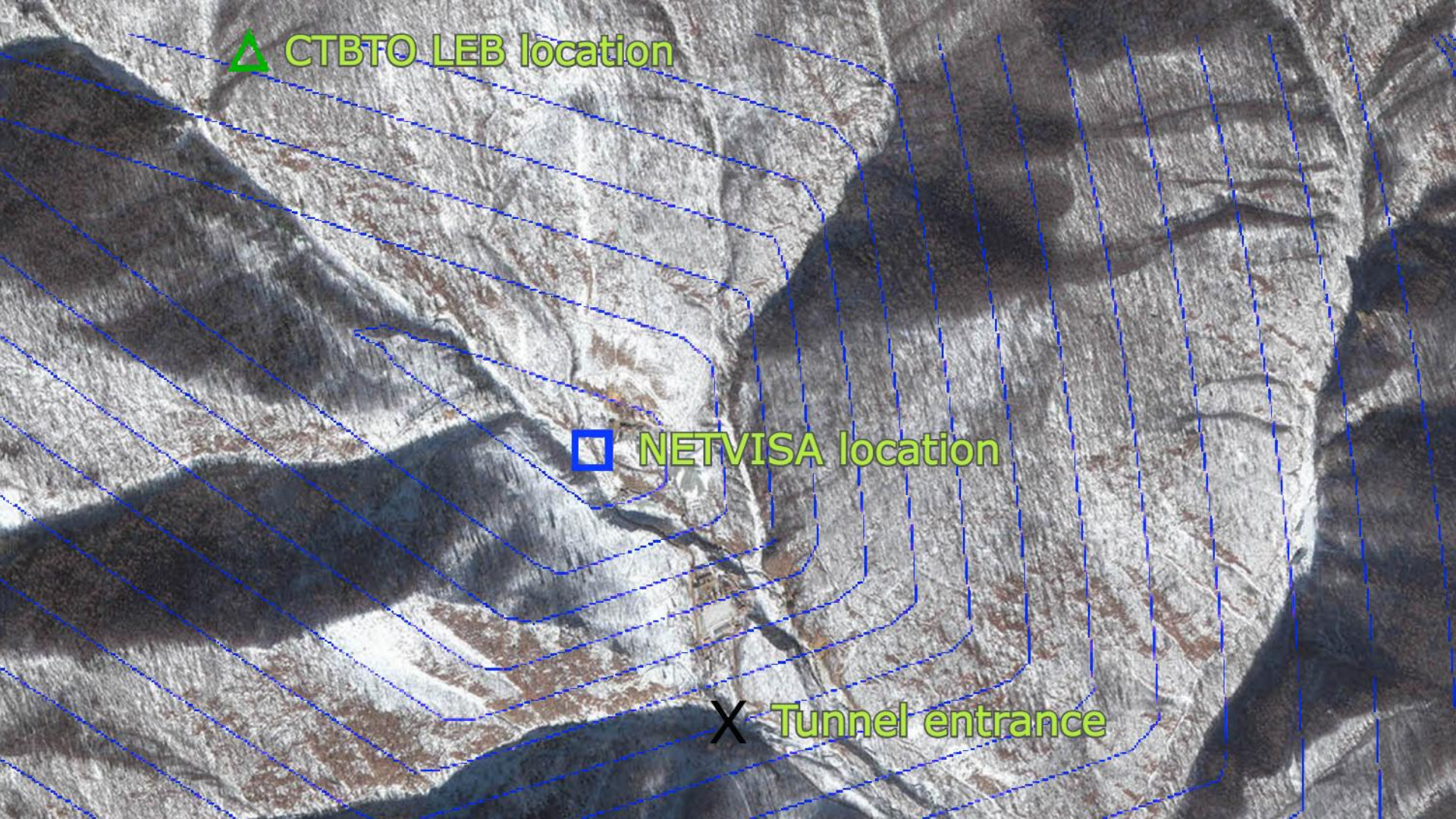
AMIT KATWALA SCIENCE FEB 16, 2022 11:00 AM

DeepMind Has Trained an AI to Control Nuclear Fusion

The Google-backed firm taught a reinforcement learning algorithm to control the fiery plasma inside a tokamak nuclear fusion reactor.



PHOTOGRAPH: CURDIN WÜTHRICH, SPC/EPFL

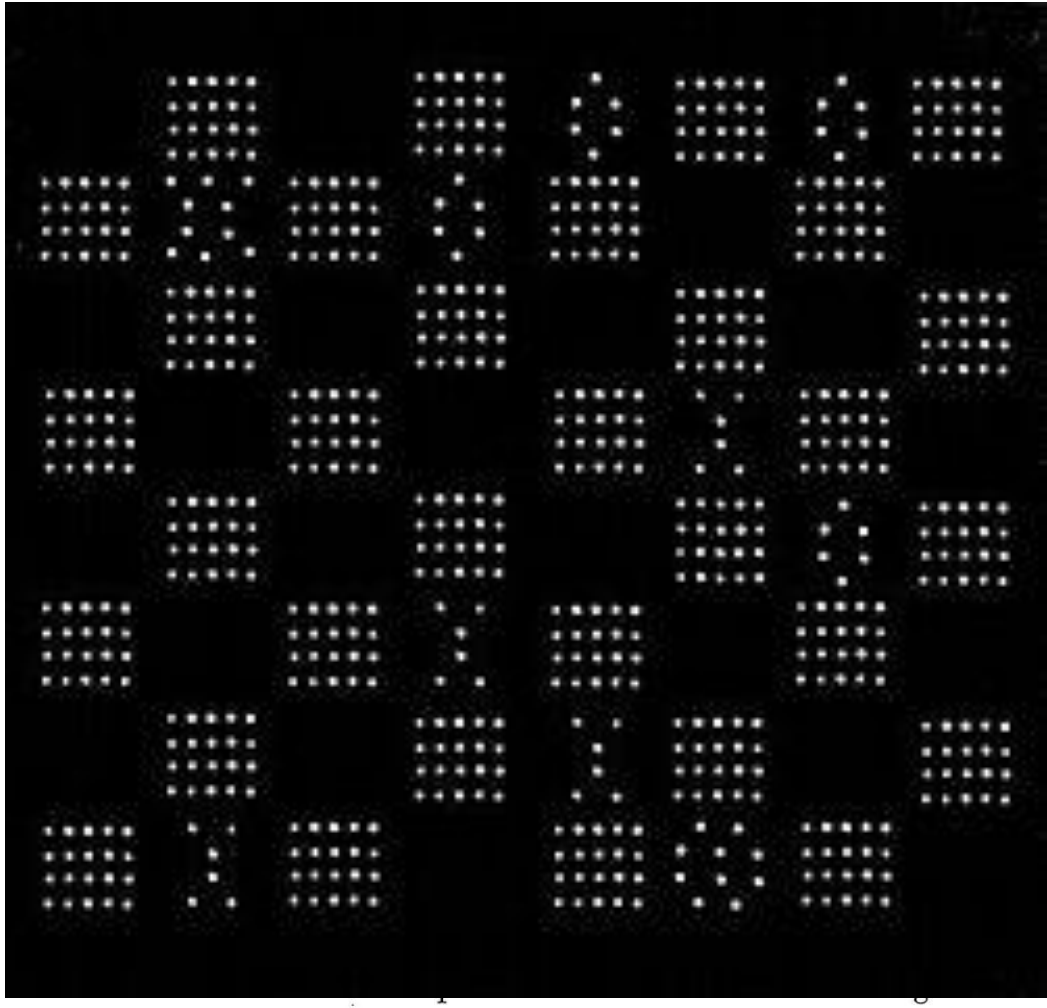

CTBTO LEB location
NETVISA location
Tunnel entrance

A short prehistory of AI

- Prehistory:
  - **Philosophy** (reasoning, planning, learning, science, automation)
  - Aristotle: For if every instrument could accomplish its own work, obeying or anticipating the will of others . . . if, in like manner, the shuttle would weave and the plectrum touch the lyre without a hand to guide them, chief workmen would not want servants, nor masters slaves
  - **Psychology** (learning, cognitive models)
  - **Linguistics** (grammars, formal representation of meaning)
- Near miss (1842):
  - Babbage design for universal machine
  - Lovelace: “a thinking machine” for “all subjects in the universe.”

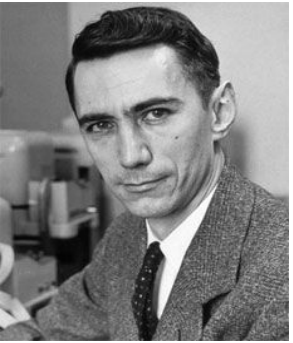
AI pre-Dartmou

- McCulloch and Pitts, 1943:
  - Perceptron/Boolean circuit model of brain
- Turing, 1950:
  - Imitation game (Turing test)
  - Machine learning as one route to AI
- Strachey, 1952:
  - Checker-playing program



AI's official birth: Dartmouth, 1956



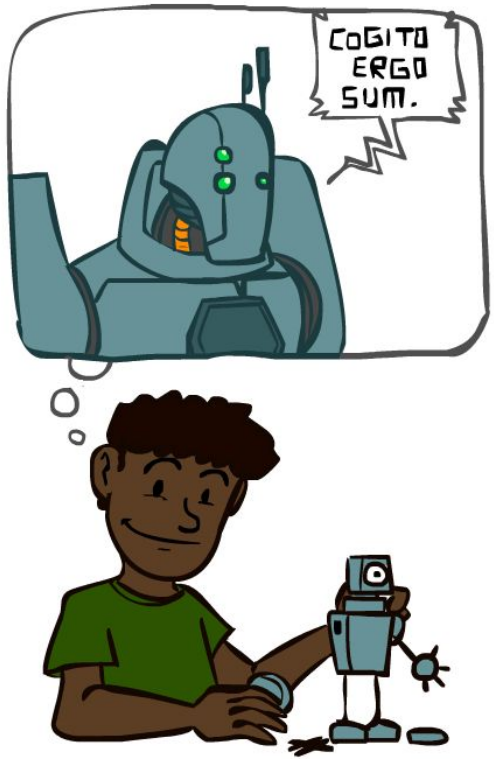


"An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer."***

John McCarthy and Claude Shannon
Dartmouth Workshop Proposal

AI post-Dartmouth

- 1956—70: Excitement: Look, Ma, no hands!
  - 1950s: Early AI programs: chess, checkers (RL), theorem proving
  - 1965: Robinson's complete algorithm for logical reasoning
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1985—90: Revival of neural nets
  - 1988—93: Expert systems industry busts: “AI Winter”
- 1990—2012: Statistical approaches + subfield expertise
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - Agents and learning systems… “AI Spring”?
- 2012—2022: Big data, deep learning, generative models
  - Image recognition, speech recognition, machine translation
  - Language models, transformers, image generators
- 2022—present: LLMs…..
  - ChatGPT, scaling laws, LLM industry booms
  - Large reasoning models

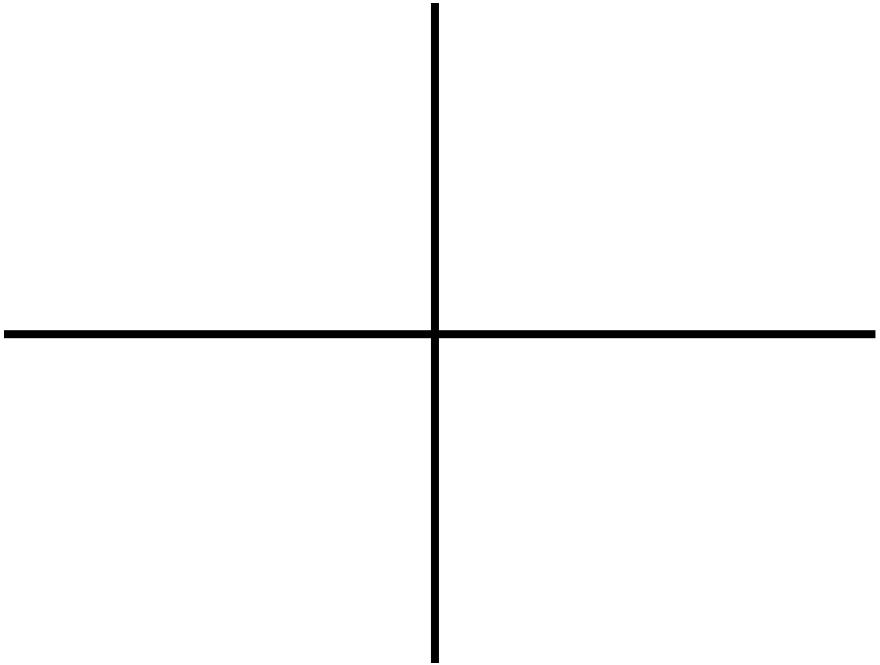


What is AI?

The attempt to understand intelligence well enough to create it in machines

And what does *that* mean?

Machines that…



What does success in AI mean?

"There are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied." Herbert Simon, 1957

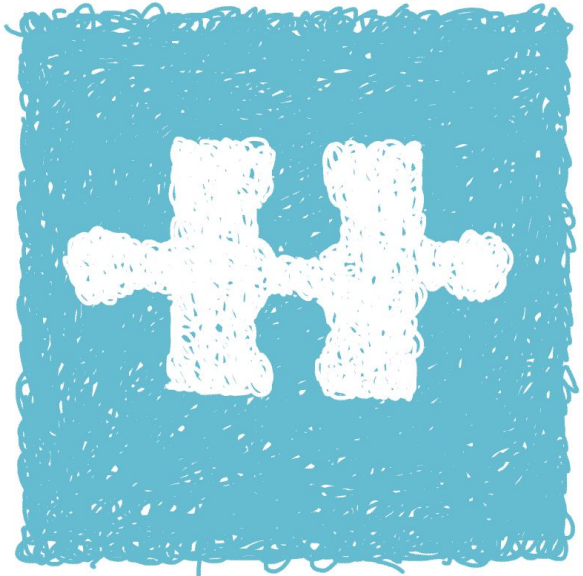


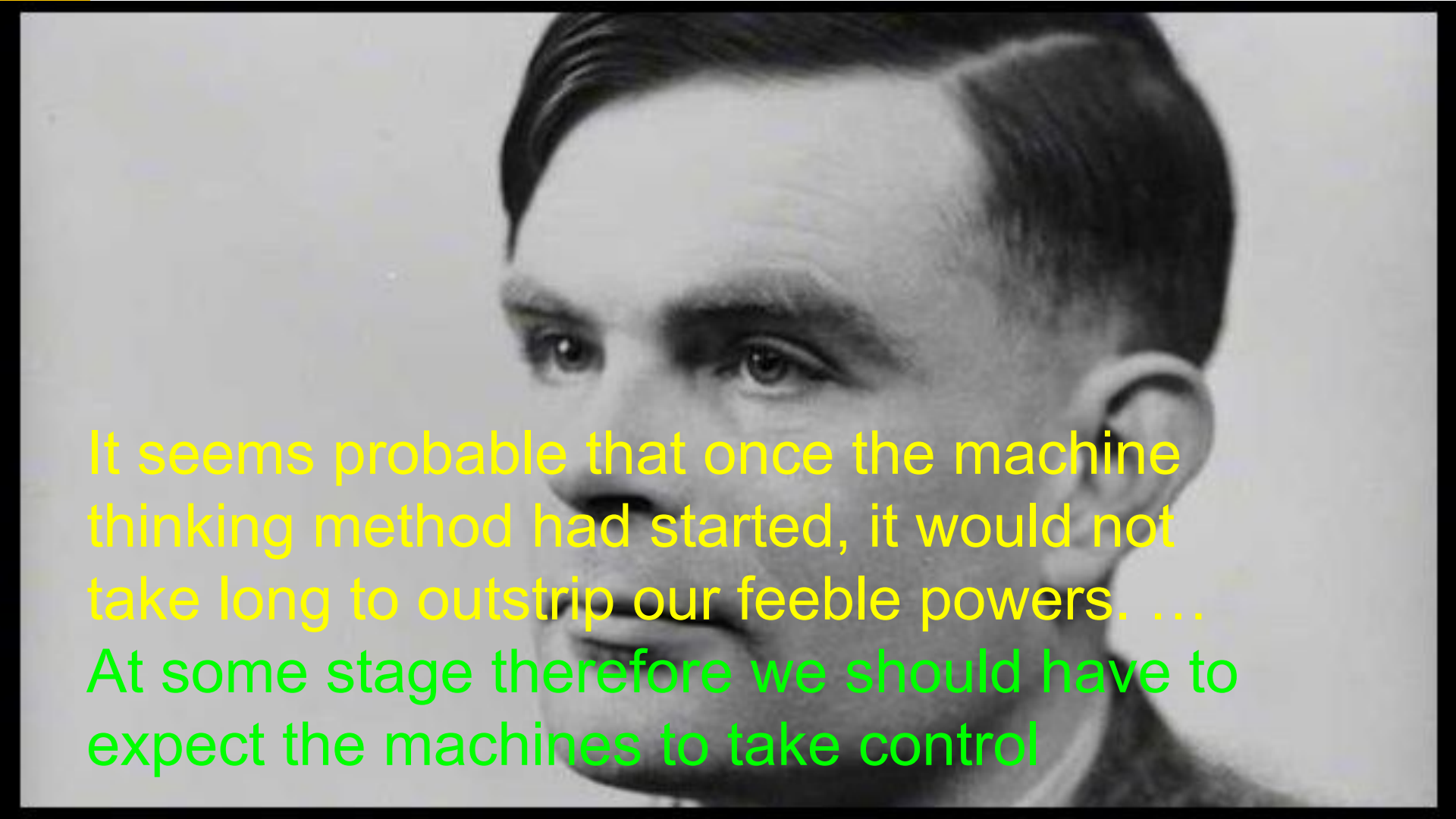
Have we succeeded?

No

GPT-like models are probably a piece of the puzzle

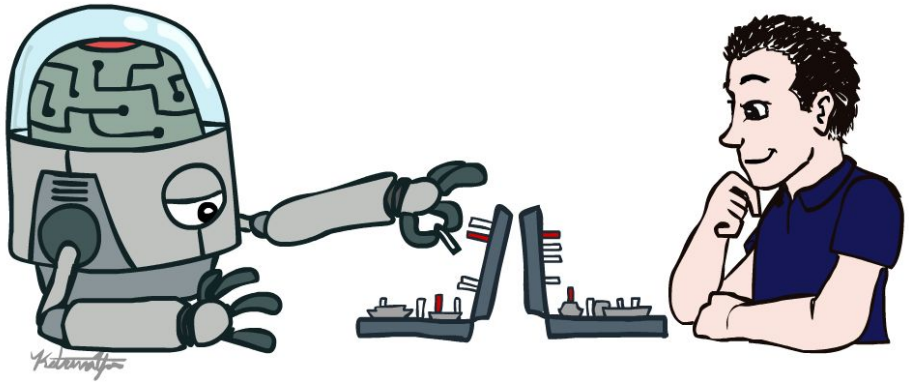
We don’t know what shape the piece is
or where it goes




It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. …
At some stage therefore we should have to expect the machines to take control

CS 188: Artificial Intelligence

Introduction Part II



Spring 2026

University of California, Berkeley

[These slides are based on those created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley (ai.berkeley.edu).]

What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
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- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along unknown city streets?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Berkeley Bowl?
- Discover and prove a new mathematical theorem?
- Perform a complex surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?



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- Win an art competition?
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- Construct a building?

Google's AlphaGo beats Lee Sedol at Go in 2016



But … a plot twist in 2023!

DEEP BLUE WAS JUST THE START —

Man beats machine at *Go* in human victory over AI

Amateur exploited weakness in systems that have otherwise dominated grandmasters.

RICHARD WATERS, FINANCIAL TIMES - 2/19/2023, 4:51 AM

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California just opened the floodgates for self-driving cars

After a state regulatory board ruling, San Francisco will have 24/7 robotaxis on its streets. A Washington Post analysis shows how it's a pivotal moment for the industry.

By Trisha Thadani and Jeremy B. Merrill

Updated August 10, 2023 at 10:06 p.m. EDT | Published August 10, 2023 at 9:15 p.m. EDT

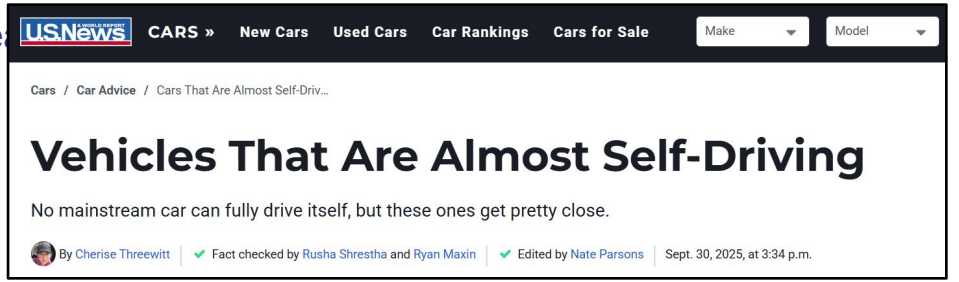


The Register

California DMV hits brakes on Cruise's SF driverless fleet after series of fender benders

50% chop effective 'immediately' as department investigates traffic and safety issues

Jude Karabus Mon 21 Aug 2023 // 14:28 UTC



U.S.News CARS » New Cars Used Cars Car Rankings Cars for Sale Make Model

Cars / Car Advice / Cars That Are Almost Self-Driv...

Vehicles That Are Almost Self-Driving

No mainstream car can fully drive itself, but these ones get pretty close.

By Cherise Threewitt | Fact checked by Rusha Shrestha and Ryan Maxin | Edited by Nate Parsons | Sept. 30, 2025, at 3:34 p.m.

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please write me a very funny, and extremely short story

A penguin walks into a bar, looks around, and says,

"Has anyone seen my brother?"

The bartender squints and asks,

"What does he look like?"

ChatGPT, 2025

Henry Squirrel was thirsty. He walked over to the river bank where his good friend Bill Bird was sitting. Henry slipped and fell in the river. Gravity drowned. The End.

Tale-Spin (Shank), 1984

What Can AI Do?

Quiz: Which of the following can be done at present?

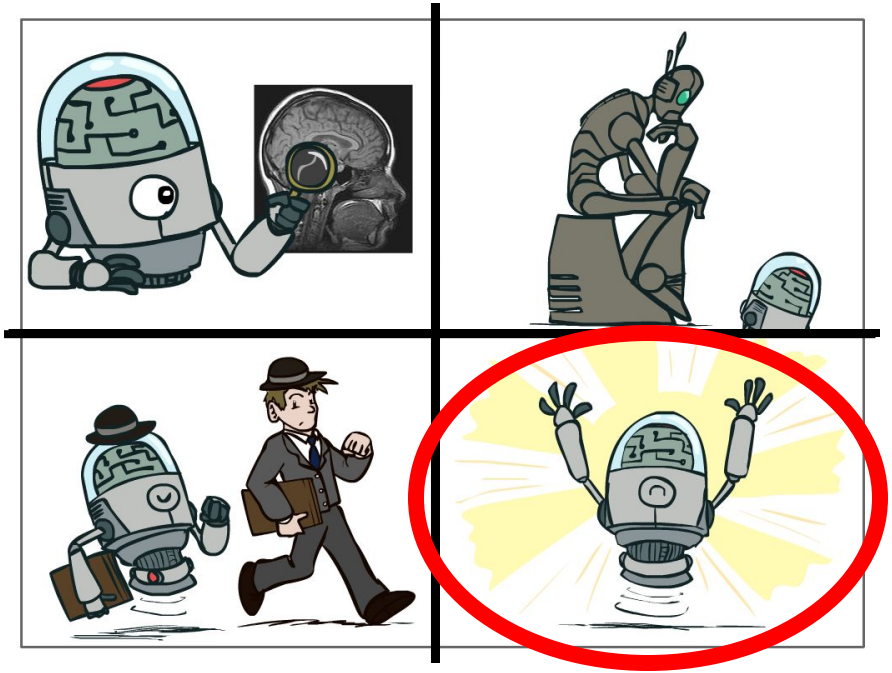
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- ✗ Construct a building?



A Rational-Agent Perspective

Should we make machines that…

Think like people?



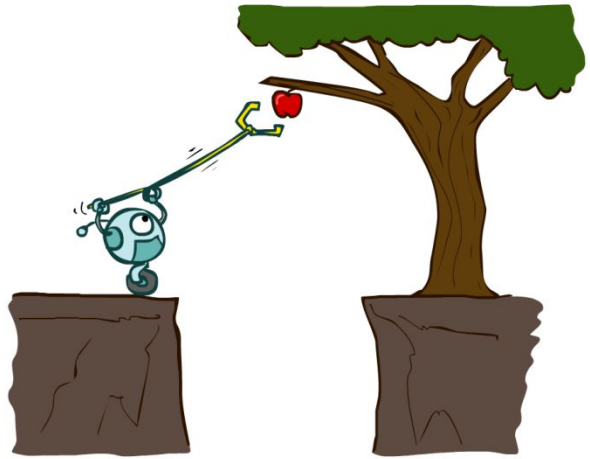
Think rationally?

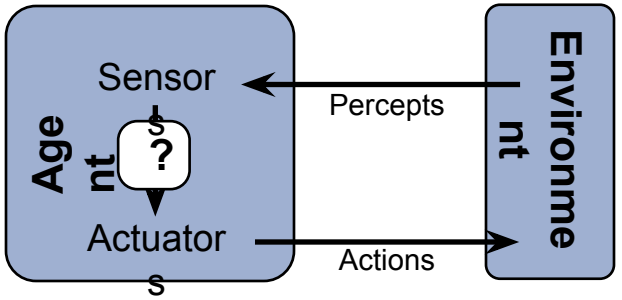
Act like people?

Act rationally?

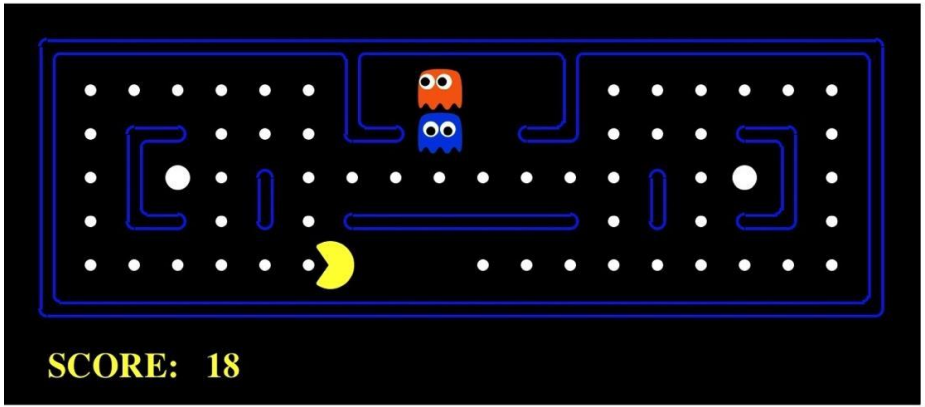
This Course: Designing Rational Agents

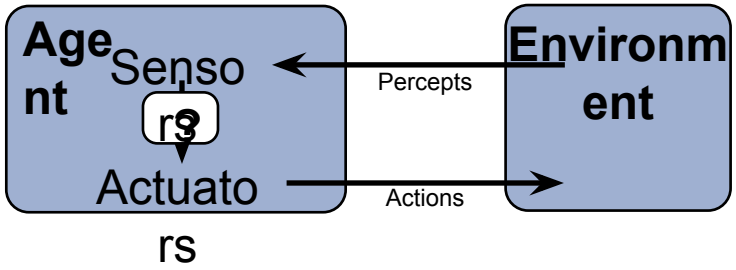
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions
- **This course** is about:
  - General AI techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique





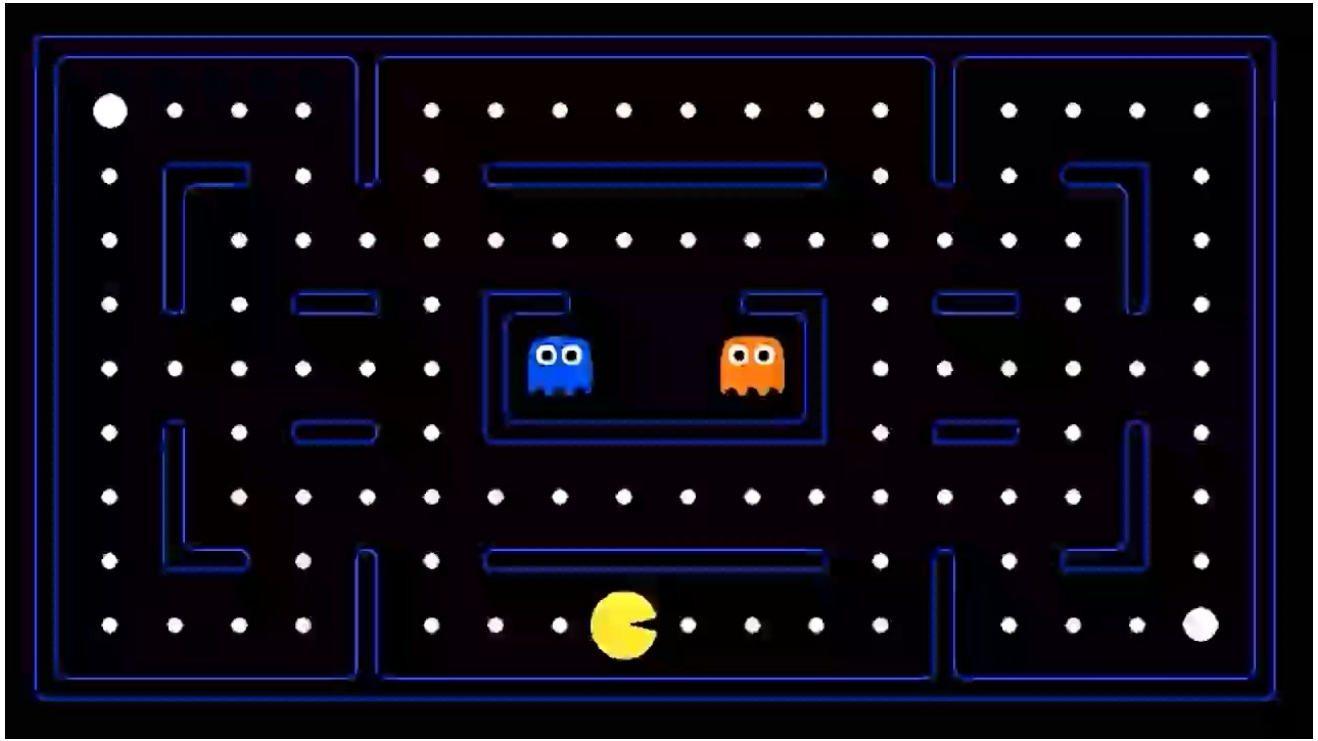
Pac-Man as an Agent



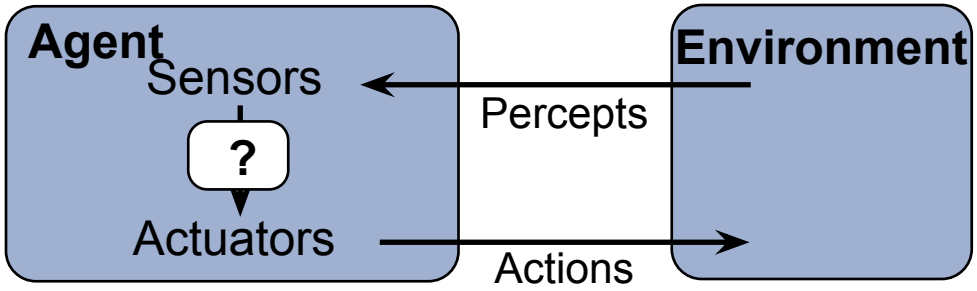


Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes

Pac-Man as an Agent



Agents and environments

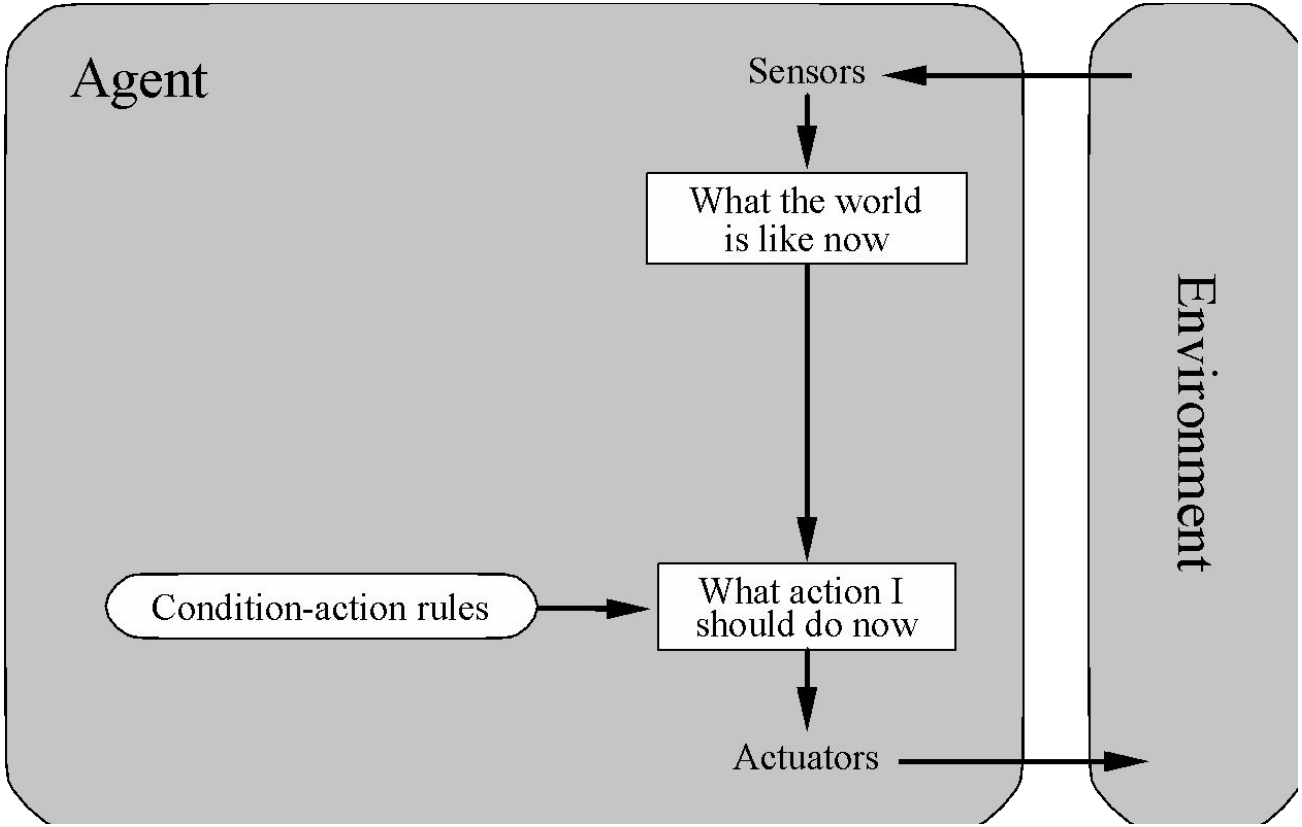


- An agent ***perceives*** its environment through ***sensors*** and ***acts*** upon it through ***actuators*** (or *effectors*, depending on whom you ask)
- The ***agent function*** maps percept sequences to actions
- It is generated by an ***agent program*** running on a ***machine***

Agent design

- **The environment type largely determines the agent design**
  - ***Partially observable*** => agent requires ***memory*** (internal state)
  - ***Stochastic*** => agent may have to prepare for ***contingencies***
  - ***Multi-agent*** => agent may need to behave ***randomly***
  - ***Static*** => agent has time to compute a rational decision
  - ***Continuous time*** => continuously operating ***controller***
  - ***Unknown physics*** => need for ***exploration***
  - ***Unknown perf. measure*** => observe/interact with ***human principal***

Simple reflex agents

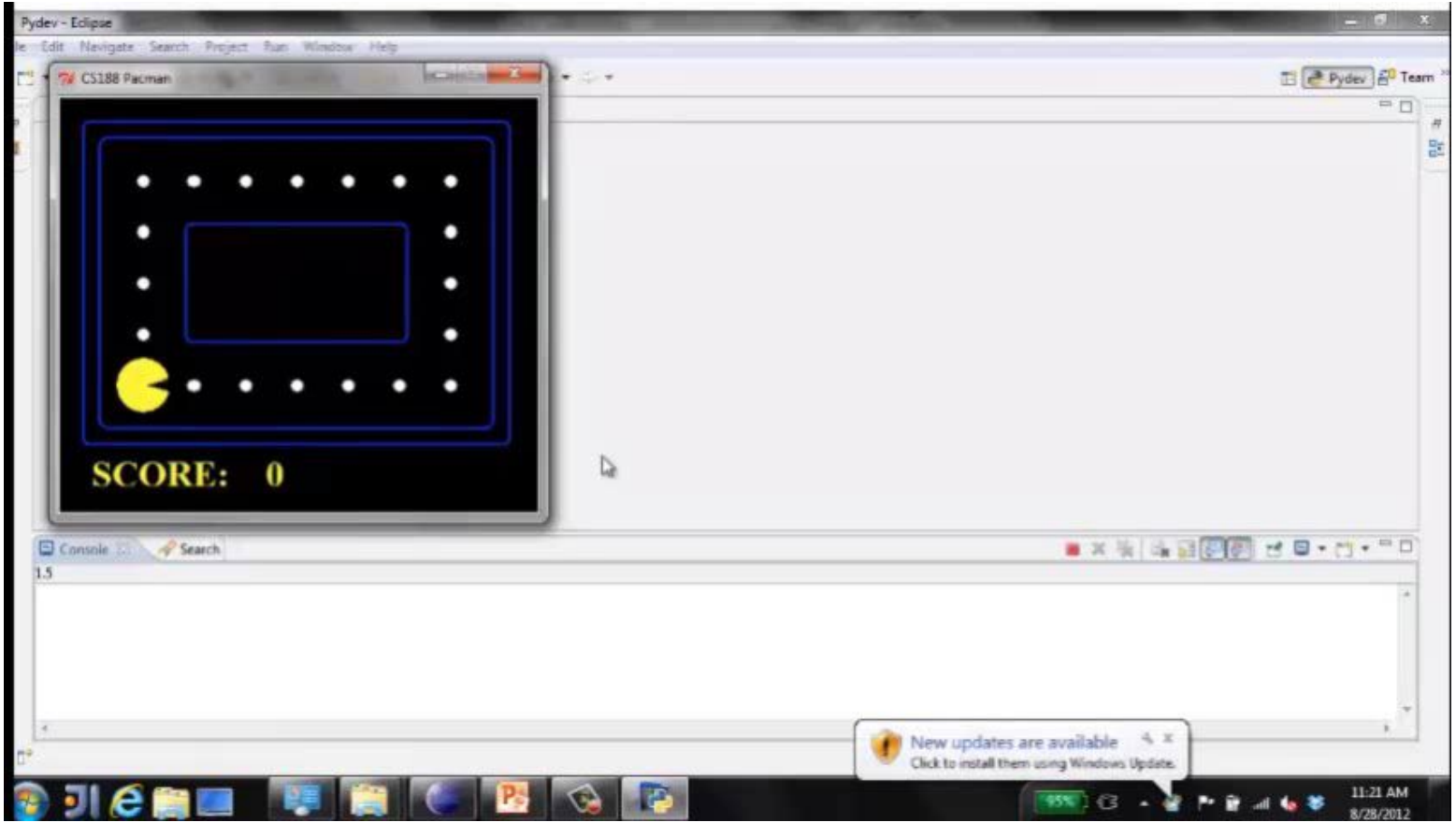


Pacman *agent program* in Python

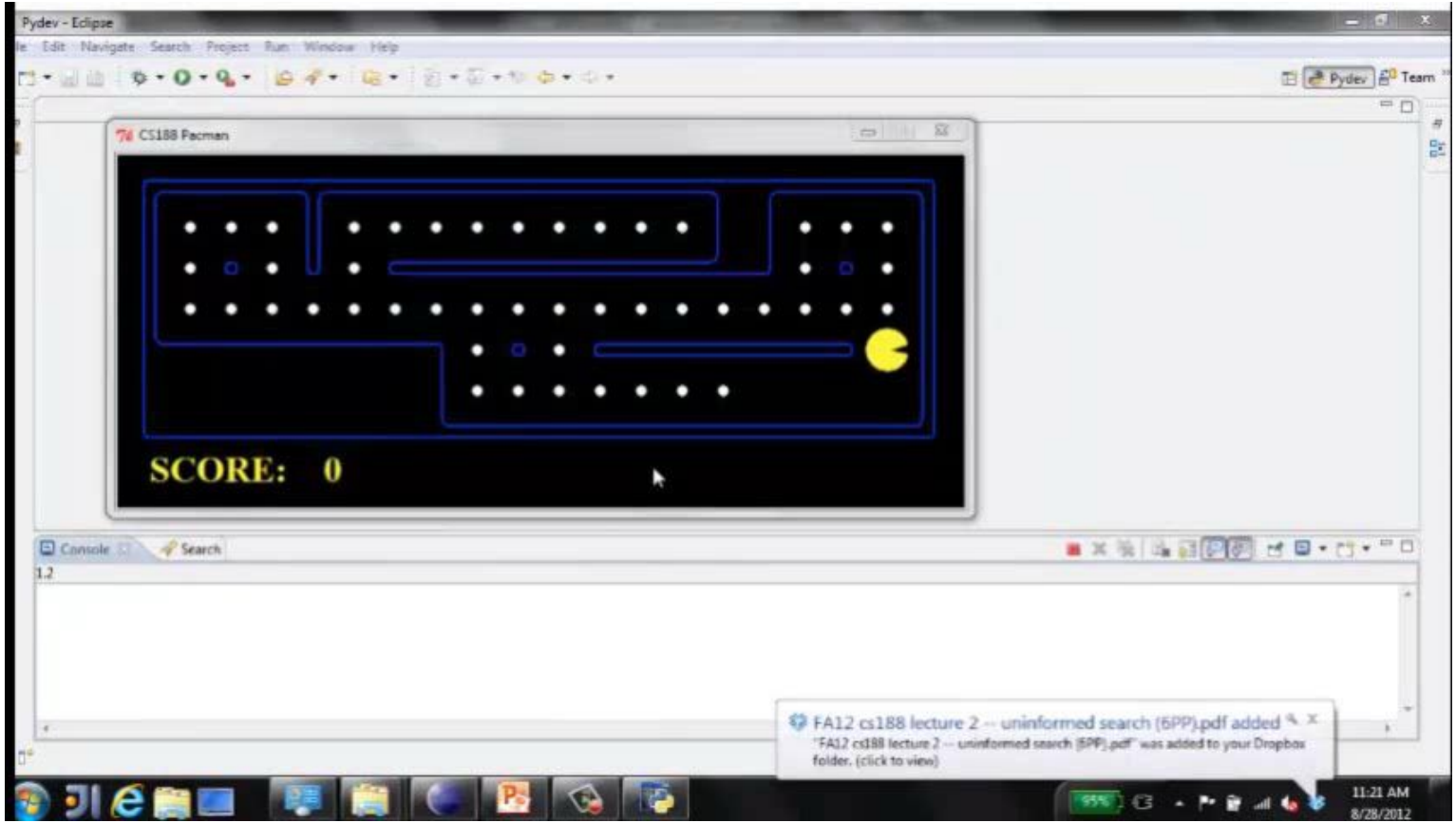
```
class GoWestAgent(Agent):

    def getAction(self, percept):
        if Directions.WEST in percept.getLegalPacmanActions():
            return Directions.WEST
        else:
            return Directions.STOP
```

Eat adjacent dot, if any



Eat adjacent dot, if any



Pacman agent contd.

- Can we (in principle) extend this reflex agent to behave well in all standard Pacman environments?
  - No – Pacman is not quite fully observable (power pellet duration)
  - Otherwise, yes – we can (*in principle*) make a lookup table…..
  - *How large would it be?*

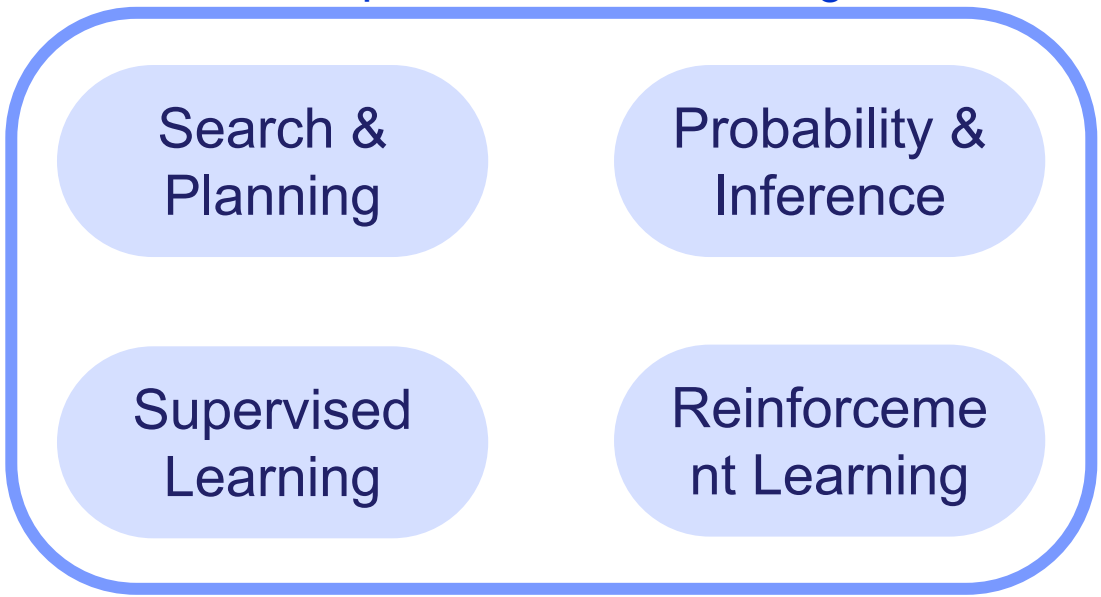
Course Topics

- Part I: Making Decisions
  - Fast search / planning
  - Constraint satisfaction
  - Adversarial and uncertain search
- Part II: Reasoning under Uncertainty
  - Probabilistic reasoning
  - Decision theory
  - Machine learning
- Part III: Applications
  - Language and LLMs, vision, robotics, games, …



Course Topics

Core Components of Rational Agents:



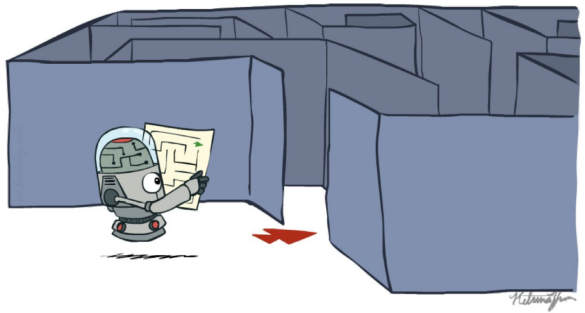
Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I use my ***model*** of the world to find a ***sequence of actions*** to achieve my ***goal***?

Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I make sense of ***uncertainty***?

Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I learn a ***model*** of the world from ***data***?

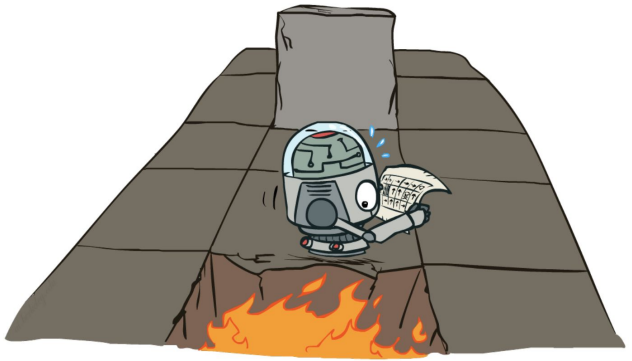
Course Topics

Search & Planning

Probability & Inference

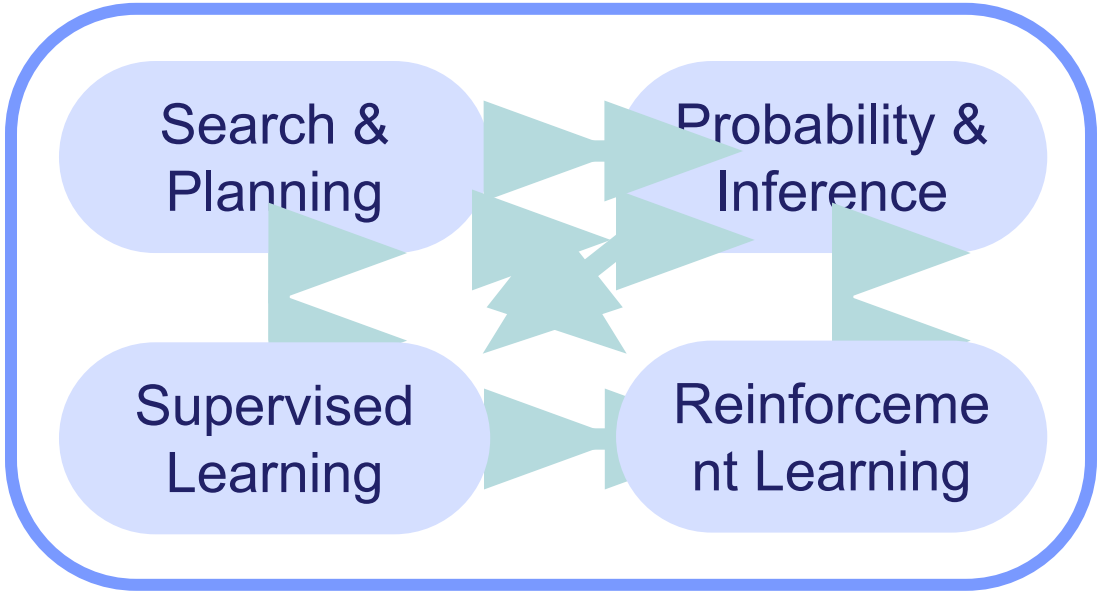
Supervised Learning

Reinforcement Learning

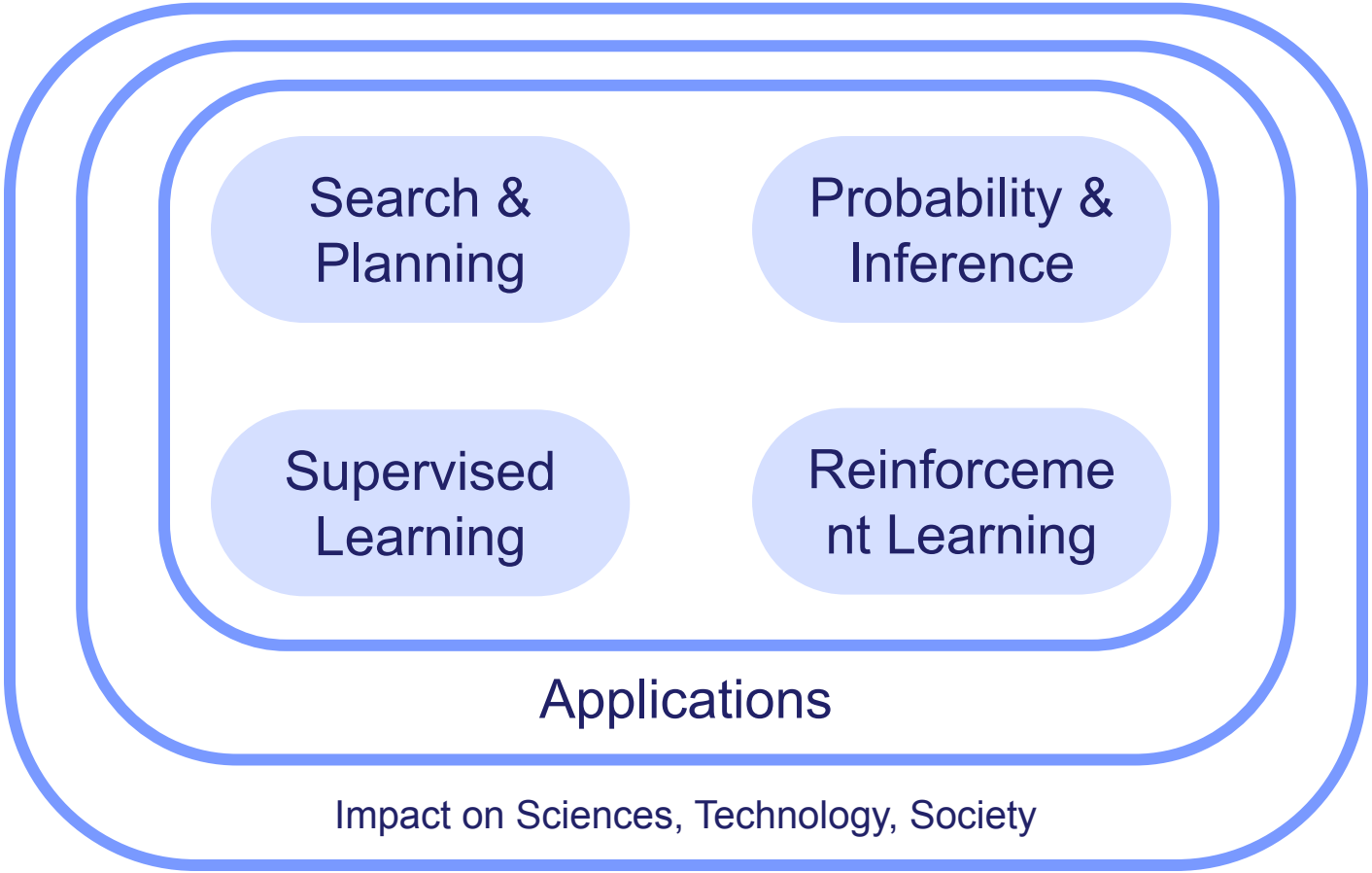


How can I learn a ***policy*** for any situation so that I can ***maximize utility***?

Course Topics



Course Topics



By the end of this course you'll:

- Build and understand math of rational, learning agents
- Select and apply the right AI methods for wide range of problems
- Recognize how these methods are used in modern AI systems
- Be prepared to make [illegible]ow AI is used in society



Next Lecture: Search

